Predictive Maintenance for Component X Failures in Heavy-Duty Scania Trucks: An Interpretable Machine Learning Approach

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I. Introduction

In today's global world, the logistics and supply chain industry has become the backbone of the economy. Within this context, the importance of vehicle reliability has increased manyfold. Because if vehicles malfunction or break unexpectedly, it leads to operational downtime and unplanned, costly repairs. So, vehicle maintenance is an important aspect of the logistics industry. In order to ensure vehicle reliability, their faults need to be identified and solved beforehand so as to avoid shutdown or delays in the supply chain network.

Traditionally, the industry relied on reactive maintenance, that is, fixing the vehicle when a fault occurs. Nevertheless, with the exponential growth of Industrial Internet of Things (IIoT), Cyber-Physical Systems (CPS), and Artificial Intelligence (AI), the industry is transitioning from Reactive Maintenance to Preventive Maintenance (PdM) [1].

These techniques use real-time and past data to predict vehicle failures, plan maintenance more effectively, and reduce unexpected delays. This significantly improves productivity, lowers costs, and increases system reliability [2].

This study lies at the intersection of industrial data mining and machine learning, and is characterized by a large sensor dataset, high class imbalance as failures are rare events, and a high demand for explainability.

Current research work often involves Deep Learning (DL) architectures like Long Short-Term Memory (LSTMs) or Transformers, as they are quite effective with sequential data, such as time-series sensor readings [3]. Nevertheless, these models operate as black boxes and thus reduce the level of trust among engineers [4].

II. Aim, objectives, and expected outcomes

A. Aim

The primary aim of this study is to train and develop a robust and interpretable statistical ML model that can effectively predict failures in SCANIA heavy-duty trucks.

B. Objectives

In order to achieve the aforementioned aim, the following measurable objectives are defined:

- **Data Preprocessing**: Build a pipeline to transform the raw SCANIA Component X dataset by converting cumulative counters and histogram data into a standard tabular format.
- **Temporal Feature Engineering**: Create new time-based features like rolling averages, variances, and trends to capture temporal dependencies in the data.
- **Imbalance Management**: Address the class imbalance problem using sampling techniques such as Synthetic Minority Over-sampling Technique (SMOTE).
- Model Optimization: Train and fine-tune gradient boosting models (XGBoost and CatBoost) to achieve the best prediction accuracy.
- **Explainability Analysis**: Apply explainability techniques (like SHAP (SHapley Additive exPlanations)) to explain the model predictions.

C. Expected outcome

The primary outcome is a robust machine learning pipeline that accurately identifies trucks that are at risk of failure with high recall. This will also demonstrate that interpretable models can effectively replace deep learning in safety-critical applications, addressing the "black box" problem.

III. Research Question

Can classical algorithms effectively mine predictive failure patterns from complex sensor data to provide an interpretable alternative to Deep Learning for truck failure predictions?

IV. Method

This study will follow the CRISP-DM (CRoss Industry Standard Process for Data Mining) methodology:

- **Business Understanding**: The primary business goal is to minimize operational and financial costs associated with unexpected truck breakdowns. This will be done by predicting the component failure beforehand and avoiding unexpected and expensive delays.
- **Data Understanding**: In this study, we will use the SCANIA Component X Dataset (2025) [5]. This industrial dataset contains operational data, truck specifications, and repair records, including cumulative counters and histogram sensor readings.
- **Data Preparation**:
  - ***Preprocessing***: Convert raw cumulative counters into time-based usage measurements and transform histograms into numerical features.
  - ***Feature Engineering***: Since deep learning methods are beyond the scope of this study, we will need to do manual feature engineering to capture temporal patterns in the data. This will be done by calculating statistical properties of the data (mean, standard deviation, slope) over sliding time windows (e.g., 7-day, 30-day) to capture the component's health trends.
- **Modeling**: The primary classifier will be a gradient boosting model (XGBoost or CatBoost). We will

also address the class imbalance by exploring various sampling techniques.

- **Evaluation**: The model will be validated using a time-aware split (training on past data, testing on future data) to prevent leakage. Performance will be assessed using Recall (Sensitivity) and F1-Score rather than accuracy. We will also explore various explainability techniques to justify the model predictions.

V. Risks and Solutions

- **Risk 1: Extreme Class Imbalance (Rare Failures)**
  - ***Severity***: High
  - ***Solution***: We will employ data sampling techniques.
- **Risk 2: High Dimensionality of Histogram Data**
  - ***Severity***: Medium
  - ***Solution***: Flattening histograms will create thousands of columns. We will use feature selection techniques to remove non-informative bins and reduce noise.
- **Risk 3: Loss of Temporal Nuance**
  - ***Severity***: Medium
  - ***Solution***: Without LSTMs, complex sequential patterns might be lost. So, we will mitigate this by performing complex feature engineering.

VI. References

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